Ember & Vine Kitchen – Data Model

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A logical and physical data model for a fictional restaurant group that starts with one dine-in restaurant and grows into franchised locations and food trucks.

Scope and design choices

The model supports daily operations (orders, inventory, labor) and strategy (location performance, menu engineering, franchise royalties) from one schema that treats a restaurant, a franchise unit and a food truck as the same thing: a **location**.

1. **One `location` entity, three types.** `location_type` is RESTAURANT, FRANCHISE or TRUCK. Trucks add a `truck_stop_schedule` for where they park; everything else (orders, labor, inventory) works the same.
2. **Franchise-ready from day one.** Every location belongs to an `operating_entity` (the company or a franchisee). Royalties are computed per entity from a `franchise_agreement`.
3. **Menu is separate from price.** `menu_item` is the product; `menu_item_price` holds price by location and date range, so franchisees and trucks can price differently.
4. **Orders are channel-agnostic.** Dine-in, takeout, delivery apps and truck window sales share one `customer_order` table, tagged by `order_channel`.
5. **Recipes link sales to inventory.** `recipe_line` ties each menu item to ingredients, so theoretical food cost and variance versus actual usage can be calculated.
6. **History is kept where it matters.** Prices and agreements use effective dates; transactions are append-only and corrected with adjustments, not edits.

Logical entity dictionary

22 entities in five subject areas. Each is identified by a surrogate primary key (PK); foreign keys (FK) carry the relationships drawn above, read as many-to-one toward the parent.

Entity	Area	PK	FKs (parent)	Key attributes
operating_entity	Organization	entity_id	none	legal_name, entity_type (COMPANY or FRANCHISEE), tax_id, created_date
franchise_agreement	Organization	agreement_id	entity_id (operating_entity)	royalty_pct, ad_fund_pct, term_start, term_end, territory_description
location	Organization	location_id	entity_id (operating_entity)	location_name, location_type (RESTAURANT, FRANCHISE, TRUCK), status, open_date, close_date, address, timezone, seat_count
truck_stop_schedule	Organization	stop_id	location_id (location, trucks only)	stop_name, latitude, longitude, start_ts, end_ts, permit_number
job_role	People	role_id	none	role_name, is_hourly, department (FOH, BOH, MGMT)
employee	People	employee_id	home_location_id (location), role_id (job_role)	first_name, last_name, hire_date, term_date, pay_type, base_rate
shift	People	shift_id	employee_id (employee), location_id (location)	scheduled_start, scheduled_end, clock_in, clock_out, hourly_rate
customer	Customers	customer_id	none	email, phone, first_name, loyalty_tier, loyalty_points, marketing_opt_in, signup_date

Entity	Area	PK	FKs (parent)	Key attributes
order_channel	Sales	channel_id	none	channel_name (DINE_IN, TAKEOUT, DELIVERY_APP, TRUCK_WINDOW, CATERING), commission_pct
customer_order	Sales	order_id	location_id (location), channel_id (order_channel), customer_id (customer, optional), server_employee_id (employee, optional)	order_ts, status, subtotal, tax, tip, discount, total, party_size
order_line	Sales	order_line_id	order_id (customer_order), menu_item_id (menu_item)	quantity, unit_price, line_discount, notes, voided_flag
payment	Sales	payment_id	order_id (customer_order)	payment_method, amount, tip_amount, processor_ref, paid_ts
menu_category	Menu	category_id	parent_category_id (menu_category, self)	category_name, display_order
menu_item	Menu	menu_item_id	category_id (menu_category)	item_name, description, is_active, is_vegetarian, is_gluten_free, launch_date
menu_item_price	Menu	price_id	menu_item_id (menu_item), location_id (location, null = chain-wide)	price, effective_from, effective_to
ingredient	Inventory	ingredient_id	none	ingredient_name, unit_of_measure, standard_cost, shelf_life_days, is_allergen
recipe_line	Inventory	recipe_line_id	menu_item_id (menu_item), ingredient_id (ingredient)	quantity_per_serving, yield_loss_pct

Entity	Area	PK	FKs (parent)	Key attributes
supplier	Inventory	supplier_id	none	supplier_name, contact_email, payment_terms_days
purchase_order	Inventory	po_id	location_id (location), supplier_id (supplier)	order_date, expected_date, received_date, status
purchase_order_line	Inventory	po_line_id	po_id (purchase_order), ingredient_id (ingredient)	quantity, unit_cost
inventory_movement	Inventory	movement_id	location_id (location), ingredient_id (ingredient)	movement_ts, movement_type (RECEIPT, USAGE, WASTE, COUNT_ADJUST, TRANSFER), quantity, source_ref
royalty_statement	Finance	statement_id	agreement_id (franchise_agreement)	period_start, period_end, gross_sales, royalty_amount, ad_fund_amount, status

Two rules keep the data clean: a `location` of type TRUCK is the only kind that may have `truck_stop_schedule` rows, and `menu_item_price.location_id` may be null only for chain-wide prices.